

PART IIB REVISION NOTES

4F7: Statistical Signal and Network Models Inference in State-Space Models

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Preliminaries

Bayes's Theorem

$$\underbrace{p(x|y, \theta)}_{\text{Posterior Distribution}} = \frac{\underbrace{p(y|x, \theta)}_{\text{Likelihood}} \underbrace{p(x|\theta)}_{\text{Prior}}}{\underbrace{p(y|\theta)}_{\text{Marginal Likelihood}}}$$

with y = observations, x = quantity of interest (state) and θ = model parameters.

$p(y|\theta) = \int_x p(y|x, \theta)p(x|\theta)dx$ is the marginal likelihood.

Multivariate Gaussian

For a length N random column vector x with mean μ and covariance C_X , the probability density function is given by:

$$f_X(x) = \frac{1}{(2\pi)^{N/2}|C_X|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^\top C_X^{-1}(x - \mu)\right)$$

The mean vector: $\mu = \mathbb{E}[X]$ and the covariance matrix: $C_X = \mathbb{E}[(X - \mu)(X - \mu)^\top]$ fully characterise the multivariate Gaussian distribution.

To sample from a multivariate Gaussian distribution:

1. Compute an M such that $C_X = MM^\top$ (e.g., via Cholesky decomposition).
2. Sample a standard normal vector $u \sim \mathcal{N}(0, I)$.
3. Transform the standard normal vector to obtain a sample from the desired distribution: $x = \mu + Mu$.
4. The resulting vector x is a random sample from $\mathcal{N}(\mu, C_X)$.

1 State-Space Models

For a time series with states x_t and observations y_t , a state-space model consists of (assuming Markov structure):

1. State evolution: $x_{t+1} \sim f(x_t|x_{t-1})$

2. Observation: $y_{t+1} \sim g(y_{t+1}|x_{t+1})$

The joint density is

$$\begin{aligned} p(x_{0:t}, y_{0:t}) &= p(x_0) \prod_{i=1}^t p(x_i|x_{0:i-1})p(y_i|x_{0:i}, y_{0:i-1}) \\ &= p(x_0) \prod_{i=1}^t f(x_i|x_{i-1})g(y_i|x_i) \end{aligned}$$

The conditional probability

$$p(x_t|y_t, x_{t-1}) = \frac{p(x_t|x_{t-1})p(y_t|x_t, x_{t-1})}{p(y_t|x_{t-1})} = \frac{f(x_t|x_{t-1})g(y_t|x_t)}{\int f(x_t|x_{t-1})g(y_t|x_t)dx_t}$$

2 Continuous-Time Models

Linear continuous-time state-space models is given by:

$$\dot{x}(t) = Ax(t) + MW(t)$$

where $W(t)$ is a continuous-time white noise process with covariance with zero mean and autocorrelation function $R_{WW}(\tau) = \sigma_W^2 \delta(\tau)$. The power of $W(t)$ is *infinite*, therefore it is often define in terms of its integral *Brownian motion* $B(t) = \int_0^t W(\tau)d\tau$. The Brownian motion follows $B(t) \sim \mathcal{N}(0, t\sigma_W^2)$.

Apply Laplace transform to both side and rearrange:

$$\bar{X}(s) = (sI - A)^{-1}x(0) + (sI - A)^{-1}M\bar{W}(s)$$

Convert back to time domain:

$$x(t) = \exp(At)x(0) + \underbrace{\int_0^t \exp(A(t-\tau))MW(\tau)d\tau}_{=C, \text{ Convolution with white noise}}$$

where $\exp(At) = I + At + A^2t^2/2! + A^3t^3/3! + \dots$

In general case, the continuouse-time model can be written in discrete-time form as

$$x(t + \delta t) = \exp(A\delta t)x(t) + C$$

In discrete form

$$x_{n+1} = \exp(A\delta t)x_n + C, C \sim \mathcal{N}(0, \text{cov}(C))$$

The covariance of C is given by:

$$\begin{aligned} \text{cov}(C) &= \mathbb{E}[CC^\top] - \overbrace{\mathbb{E}[C]\mathbb{E}[C]^\top}^{=0} \\ &= \int_0^t \int_0^t \exp(A(t-\tau))MR_{WW}(\tau-\tau')M^\top \exp(A^\top(t-\tau'))d\tau d\tau' \\ &= \sigma_W^2 \int_0^t \exp(A(t-\tau))MM^\top \exp(A^\top(t-\tau))d\tau \end{aligned}$$

For the Ornstein-Uhlenbeck (OU) process with $A = \rho$ and $M = b$, the covariance is $\text{cov}(C) = -\frac{\sigma_W^2 b^2}{2\rho}(1 - \exp(2\rho t))$, giving $x(t + \delta t) \sim \mathcal{N}\left(\exp(\rho\delta t)x(t), -\frac{\sigma_W^2 b^2}{2\rho}(1 - \exp(2\rho\delta t))\right)$. The Euler approximation is $x(t + \delta t) \sim \mathcal{N}(x(t) + \rho x(t)\delta t, b^2\delta t\sigma_W^2) = \text{first order expansion}$.

3 Filtering

Given $p(x_t|y_{1:t})$, find $p(x_{t+1}|y_{1:t+1})$ using filtering recursions in two steps: Prediction and Correction.

Prediction step ('Chapman-Kolmogorov')

$$\begin{aligned} p(x_{t+1}|y_{1:t}) &= \int p(x_t, x_{t+1}|y_{1:t})dx_t \\ &= \int p(x_t|y_{1:t})p(x_{t+1}|x_t, y_{1:t})dx_t \\ &= \int p(x_t|y_{1:t})f(x_{t+1}|x_t)dx_t \end{aligned}$$

Correction step (Bayes's Theorem)

$$\begin{aligned} p(x_{t+1}|y_{1:t+1}) &= \frac{p(y_{t+1}|x_{t+1}, y_{1:t})p(x_{t+1}|y_{1:t})}{p(y_{t+1}|y_{1:t})} \\ &= \frac{g(y_{t+1}|x_{t+1})p(x_{t+1}|y_{1:t})}{p(y_{t+1}|y_{1:t})} \end{aligned}$$

4 Kalman Filter

Kalman filter can be applied to linear Gaussian state-space models, which follows:

$$\begin{aligned} f(x_t|x_{t-1}) &= \mathcal{N}(Ax_{t-1}, \Sigma_v) \longrightarrow x_t = Ax_{t-1} + v_t \\ g(y_t|x_t) &= \mathcal{N}(Bx_t, \Sigma_w) \longrightarrow y_t = Bx_t + w_t \end{aligned}$$

where v_t and w_t are independent Gaussian noises with zero mean and covariance Σ_v and Σ_w respectively. The initial state is also Gaussian distributed: $f(x_0) = \mathcal{N}(\mu_{0|0}, P_{0|0})$.

Goal: Find the update rules to the filtering distribution $p(x_t|y_{1:t}) = \mathcal{N}(\mu_{t|t}, P_{t|t})$.

Kalman Filter for General Case

$$\begin{aligned} \mu_{t|t-1} &= A\mu_{t-1|t-1} \\ P_{t|t-1} &= AP_{t-1|t-1}A^\top + \Sigma_v \\ K_t &= P_{t|t-1}B^\top(BP_{t|t-1}B^\top + \Sigma_w)^{-1} \\ P_{t|t} &= (I - K_tB)P_{t|t-1} \\ \mu_{t|t} &= \mu_{t|t-1} + K_t(y_t - B\mu_{t|t-1}) \end{aligned}$$

- Kalman Gain K_t : How much the prediction should be adjusted based on the new observation.
- Innovation $y_t - B\mu_{t|t-1}$: How different is the observation from the prediction.

When the state space model is linear and Gaussian (v_t, w_t, x_0 are Gaussian distributed), Kalman filter is the *optimal Bayesian estimator* of the state.

More generally, Kalman filter is the *best linear, unbiased estimate (BLUE)* in an *minimum mean square error (MMSE)* sense of the state given that v_t, w_t are zero mean and uncorrelated.

Prediction Error Decomposition with Kalman Filter

The full likelihood is given by:

$$p(y_{1:T}) = \prod_{t=1}^{T-1} p(y_{t+1}|y_{1:t})$$

where $p(y_{t+1}|y_{1:t}) = \mathcal{N}(B\mu_{t+1|t}, BP_{t+1|t}B^\top + \Sigma_w)$, subject to a parameter of interest θ .

Maximum likelihood estimate: $\hat{\theta}_{\text{ML}} = \arg \max_{\theta} p(y_{1:T}|\theta)$

Maximum a posteriori estimate: $\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(y_{1:T}|\theta)p(\theta)$

5 The Extended Kalman Filter

Non-linear state-space model:

$$\begin{aligned} x_t &= A(x_{t-1}, v_t) \\ y_t &= B(x_t, w_t) \end{aligned}$$

where $v_t \sim \mathcal{N}(0, \Sigma_v)$ and $w_t \sim \mathcal{N}(0, \Sigma_w)$, can be linearised by expanding A and B using 1st order multivariate Taylor expansion around $x_t = x^*, v_t = 0$ and $x_t = x^+, w_t = 0$ respectively:

$$\begin{aligned} A(x_{t-1}, v_t) &\approx A(x^*, 0) + \frac{\partial A}{\partial x_{t-1}}(x_{t-1} - x^*) + \frac{\partial A}{\partial v_t}v_t = A'x_{t-1} + C + Dv_t \\ B(x_t, w_t) &\approx B(x^+, 0) + \frac{\partial B}{\partial x_t}(x_t - x^+) + \frac{\partial B}{\partial w_t}w_t = B'x_t + E + Fw_t \end{aligned}$$

Usually, the linearisation points are $x^* = \mu_{t-1|t-1}$ and $x^+ = \mu_{t|t-1}$.

The Extended Kalman Filter:

$$\begin{aligned} \mu_{t|t-1} &= A'\mu_{t-1|t-1} + C \\ P_{t|t-1} &= A'P_{t-1|t-1}A'^\top + D\Sigma_vD^\top \\ K_t &= P_{t|t-1}B'^\top(B'P_{t|t-1}B'^\top + F\Sigma_wF^\top)^{-1} \\ P_{t|t} &= (I - K_tB')P_{t|t-1} \\ \mu_{t|t} &= \mu_{t|t-1} + K_t(y_t - B'\mu_{t|t-1} - E) \end{aligned}$$

6 Particle Filters

General Monte Carlo

Use random samples to represent the distribution:

$$x^{(i)} \stackrel{\text{i.i.d.}}{\sim} p(x), \quad i = 1, \dots, N \text{ for large } N$$
$$p(x) \approx \frac{1}{N} \sum_{i=1}^N \underbrace{\delta(x - x^{(i)})}_{\delta_{x^{(i)}}(x)}$$

Monte Carlo Expectation:

$$\bar{h} = \int h(x)p(x)dx \approx \int h(x)\frac{1}{N} \sum_{i=1}^N \delta_{x^{(i)}}(x)dx \approx \frac{1}{N} \sum_{i=1}^N h(x^{(i)})$$

Importance Sampling

Draw random samples from a distribution $q(x)$ instead of $p(x)$, the expectation with respect to $p(x)$ can be estimated with

$$\bar{h} = \int h(x)\frac{p(x)}{q(x)}q(x)dx \approx \frac{1}{N} \sum_{i=1}^N h(x^{(i)}) \underbrace{\frac{p(x^{(i)})}{q(x^{(i)})}}_{\tilde{w}^{(i)}}, \quad x^{(i)} \stackrel{\text{i.i.d.}}{\sim} q(x)$$

where $\tilde{w}^{(i)} = \frac{p(x^{(i)})}{q(x^{(i)})}$ is the unnormalised importance weight.

Monte Carlo Filtering

Infer the filtering distribution $p(x_t|y_{1:t})$ or estimate the expectation with importance sampling:

$$\bar{h} = \int h(x_t)p(x_t|y_{1:t})dx_t \approx \frac{1}{N} \sum_{i=1}^N w_t^{(i)} h(x_t^{(i)}), \quad x_t^{(i)} \stackrel{\text{i.i.d.}}{\sim} q(x_t)$$

where $w_t^{(i)} \propto \frac{p(x_t^{(i)}|y_{1:t})}{q(x_t^{(i)})}$ is the normalised importance weight. The filtering density is approxiamted as $p(x_t|y_{1:t}) \approx \sum_{i=1}^N w_t^{(i)} \delta_{x_t^{(i)}}(x_t)$.

The normalisation constant is a Monte Carlo estimate $\hat{Z} = \sum_{i=1}^N \tilde{w}^{(i)}$ and $w^{(i)} = \frac{\tilde{w}^{(i)}}{\hat{Z}}$.

Resampling

For $i = 1, \dots, N$, set $x'^{(i)} = x^{(j)}$ with probability $w^{(j)}$, i.e. each $x'^{(i)}$ is drawn from a categorical distribution with probabilities $w^{(1)}, \dots, w^{(N)}$.

The density function $p(x)$ transform from a weighted to an unweighted approximation:

$$p(x) \approx \sum_{i=1}^N w^{(i)} \delta_{x^{(i)}}(x) \xrightarrow{\text{Resampling}} p(x) \approx \frac{1}{N} \sum_{i=1}^N \delta_{x'^{(i)}}(x)$$

Benefits: limit the degeneracy of the importance weights over time – all probability mass becomes concentrated on a few particles, which makes the estimates unreliable.

Sequential Monte Carlo

Given a collection of samples ('particles'), the filtering density is approximated as

$$p(x_t|y_{1:t}) \approx \sum_{i=1}^N \delta_{x_t^{(i)}}(x_t), \quad x_t^{(i)} \stackrel{\text{i.i.d.}}{\sim} p(x_t|y_{1:t}), \quad i = 1, \dots, N$$

Prediction step:

$$\begin{aligned} p(x_{t+1}|y_{1:t}) &= \int f(x_{t+1}|x_t) p(x_t|y_{1:t}) dx_t \\ &\approx \int f(x_{t+1}|x_t) \frac{1}{N} \sum_{i=1}^N \delta_{x_t^{(i)}}(x_t) dx_t \\ &= \frac{1}{N} \sum_{i=1}^N f(x_{t+1}|x_t^{(i)}) \end{aligned}$$

Update step:

$$p(x_{t+1}|y_{1:t+1}) = \frac{g(y_{t+1}|x_{t+1}) p(x_{t+1}|y_{1:t})}{p(y_{t+1}|y_{1:t})} \approx \frac{1}{N} \frac{g(y_{t+1}|x_{t+1}) \sum_{i=1}^N f(x_{t+1}|x_t^{(i)})}{p(y_{t+1}|y_{1:t})}$$

Samples can be drawn from the Monte Carlo approximation to $p(x_{t+1}|y_{1:t+1})$ by:

- rejection sampling (slow but exact)
- importance sampling (most common)
- MCMC (effective but slow)
- annealed importance sampling etc.

General Sequential Importance Sampling

Generate a new state from a specially designed proposal distribution $q(x_t, x_{t+1}|y_{1:t+1})$ rather than the state transition density $f(x_{t+1}|x_t)$:

$$\begin{aligned} q(x_t, x_{t+1}|y_{1:t+1}) &= q(x_{t+1}|x_t, y_{1:t+1}) p(x_t|y_{1:t}) \\ p(x_t, x_{t+1}|y_{1:t+1}) &= \frac{g(y_{t+1}|x_{t+1}) f(x_{t+1}|x_t) p(x_t|y_{1:t})}{p(y_{t+1}|y_{1:t})} \end{aligned}$$

The importance weight is given by:

$$\begin{aligned} w_{t+1} &\propto \frac{p(x_t, x_{t+1}|y_{1:t+1})}{q(x_t, x_{t+1}|y_{1:t+1})} \\ &= \frac{g(y_{t+1}|x_{t+1}) f(x_{t+1}|x_t) p(x_t|y_{1:t})}{p(y_{t+1}|y_{1:t}) q(x_{t+1}|x_t, y_{1:t+1}) p(x_t|y_{1:t})} \\ &\propto \frac{g(y_{t+1}|x_{t+1}) f(x_{t+1}|x_t)}{q(x_{t+1}|x_t, y_{1:t+1})} \end{aligned}$$

If resampling is not carried out, weights are accumulated over time:

$$w_{t+1} \propto w_t \frac{g(y_{t+1}|x_{t+1})f(x_{t+1}|x_t)}{q(x_{t+1}|x_t, y_{1:t+1})}$$

Otherwise, the weights are reset to $1/N$ after resampling.

Important special cases:

- Bootstrap filter: $q(x_{t+1}|x_t, y_{1:t+1}) = f(x_{t+1}|x_t)$
- Sequential computations: $q(x_{t+1}|x_t, y_{1:t+1}) = p(x_{t+1}|x_t, y_{t+1})$

Algorithm 1 General Particle Filter

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for  $t = 1, \dots, T$  do
  for  $i = 1, \dots, N$  do
     $x_{t+1}^{(i)} \sim q(x_{t+1}^{(i)}|x_t^{(i)}, y_{1:t+1})$ 
     $w_{t+1}^{(i)} \propto w_t^{(i)} \frac{g(y_{t+1}|x_{t+1}^{(i)})f(x_{t+1}^{(i)}|x_t^{(i)})}{q(x_{t+1}^{(i)}|x_t^{(i)}, y_{1:t+1})}$ 
  end for
  (Optional Resample):
  Normalise weights,  $\sum_i w_{t+1}^{(i)} = 1$ 
  for  $i = 1, \dots, N$  do
    Set  $x_{t+1}'^{(i)} = x_{t+1}^{(j)}$  with probability  $w_{t+1}^{(j)}$ 
  end for
  for  $i = 1, \dots, N$  do
    Set  $x_{t+1}^{(i)} = x_{t+1}'^{(i)}$  and  $w_{t+1}^{(i)} = 1/N$ 
  end for
end for

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